Handling Data 7 — Compound probability · Tree diagrams · Conditional probability

Marks in brackets. ★ = harder question. Without replacement, the second probability uses one fewer item.

1 A box of 12 chocolates contains 5 dark, 4 milk and 3 white. Two chocolates are taken at random without replacement. Find the probability that [9]

a both are the same type **b** at least one is dark **c** exactly one is milk.

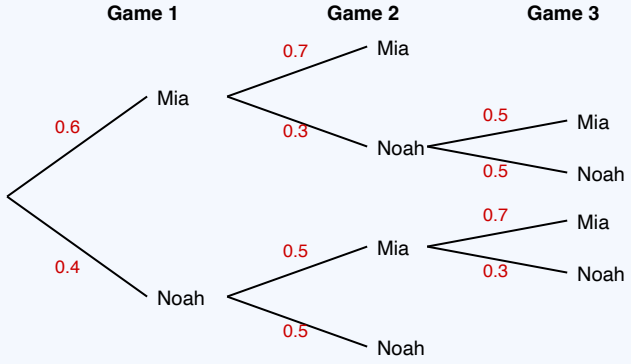
a $P(\text{DD}) + P(\text{MM}) + P(\text{WW}) = \dfrac{5}{12}\times\dfrac{4}{11} + \dfrac{4}{12}\times\dfrac{3}{11} + \dfrac{3}{12}\times\dfrac{2}{11}$, which is $\dfrac{20+12+6}{132} = \dfrac{38}{132} = \dfrac{19}{66}$.

b $1 - P(\text{no dark}) = 1 - \dfrac{7}{12}\times\dfrac{6}{11} = 1 - \dfrac{42}{132} = \dfrac{90}{132} = \dfrac{15}{22}$.

c Milk then not, or not then milk: $\dfrac{4}{12}\times\dfrac{8}{11} + \dfrac{8}{12}\times\dfrac{4}{11} = \dfrac{64}{132} = \dfrac{16}{33}$.

2 Mia and Noah play games of chess. The probability that Mia wins the first game is 0.6. When Mia wins a game, the probability that she wins the next is 0.7; when Noah wins a game, the probability that he wins the next is 0.5. The first to win two games wins the match. **a** Draw a tree diagram for the match. **b** Find the probability that Mia wins the match. [7]

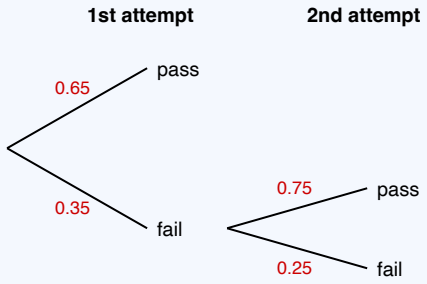
a The match ends as soon as someone has two wins, so branches stop after MM and NN.



b Mia wins via MM, MNM or NMM:
$0.6\times0.7 + 0.6\times0.3\times0.5 + 0.4\times0.5\times0.7 = 0.42 + 0.09 + 0.14 = 0.65$.

3 Sofia is taking her driving test. The probability that she passes at the first attempt is 0.65. If she fails, the probability that she passes at the next attempt is 0.75, and the same at every later attempt. Once she passes she stops. **a** Draw a tree diagram for her first two attempts. **b** Find the probability that she needs exactly two attempts. **c** Find the probability that she passes at her third or fourth attempt. [9]

a



b Fail then pass: $0.35 \times 0.75 = 0.2625$.

c Third attempt: $0.35 \times 0.25 \times 0.75 = 0.065625$. Fourth: $0.35 \times 0.25 \times 0.25 \times 0.75 = 0.0164$. Total 0.0820 (3 s.f.).

Total: 25 marks

Check: without replacement, the denominator drops by 1 and the numerator too if the same type was taken · "at least one" $= 1 - P(\text{none})$ · "exactly one" has two orders · a match or test tree stops when the outcome is decided · multiply along, add across.